INTERNSHIP REPORT

Submitted by

MOHAMED ASLAM M (21IT066)

BACHELOR OF ENGINEERING

in

INFORMATION TECHNOLOGY

THIAGARAJAR COLLEGE OF ENGINEERING,
MADURAI – 625 015

DECEMBER 18 2023 – DECEMBER 31 2023

THIAGARAJAR COLLEGE OF ENGINEERING(TCE),
MADURAI- 625 015

BONAFIDE CERTIFICATE

Certified that this Internship report for the domain of **Data Science** is the bonafide work of **MOHAMED ASLAM M** from the Department of **INFORMATION TECHNOLOGY** who carried out the Internship at Yalimart between **DECEMBER 18 2023-DECEMBER 31 2023.**

Submitted for Evaluation held at Thiagarajar College of Engineering on January 2024.

EXAMINER 1	**EXAMINER 2**
(Name with Signature)	**(Name with Signature)**

Table of Contents

Chapter No	**Title**	**Page No**
	Bonafide Certificate	1
	Abstract	2
1.	Introduction	3
2.	Background	3
3.	Objectives	5
4.	Purpose of the Work	6
5.	Concepts Explored	7
6.	Detailed Description of Learning	7
7.	Results and Discussion	13
8.	Conclusion and Future Enhancements	14
9.	References	14

CERTIFICATE:

Yalimart

Delivering Your Demand At Your Doorstep

Hig 45, 80 Feet Road, Anna Nagar, Madurai – 625020

Internship Certificate

This is to bring to your kind notice that **Mr. Mohamed Aslam M** pursuing **Bachelor of Information Technology** from **Thiagarajar College of Engineering,** has successfully completed his Internship under the guidance of Mr. Saravana Kumar G (Talent Acquisition-HR) at **Yalimart** from **18 December 2023 to 31 December 2023**.

Topic: Data Analyst

We found him Sincere, Hardworking, Dedicated and result Oriented. He worked well as part of the Team during his Tenure. We take this Opportunity to Thank him and wish him All the Best for his Future.

This Certificate was Awarded by:

Saravana Kumar G	**31.12.2023**
Human Resources	**Date**

+91 86672 12523 yalimartindia@gmail.com www.yalimart.com

Abstract

This internship focused on cultivating a comprehensive skill set in Data Science, emphasizing a carefully selected technology stack that includes Python, TensorFlow, Pandas, NumPy, and SQL. The program aimed to equip the intern with practical proficiencies essential for contemporary data analysis and machine learning applications. The journey commenced with Python, delving into the creation of data-driven solutions and exploratory data analysis. Proficiencies in data manipulation, statistical analysis, and visualization were refined, laying a solid foundation for working with large datasets. Transitioning to machine learning, dedicated efforts were invested in TensorFlow—a powerful framework. The focus included building and training machine learning models, implementing neural networks, and leveraging TensorFlow's features for efficient model deployment. Significant attention was given to data storage and retrieval using SQL, enhancing skills in designing well-structured databases and crafting optimized queries for data extraction and analysis.

Concurrently, principles of data visualization and interpretation were deepened, utilizing tools like Matplotlib and Seaborn. The internship also encompassed collaborative practices, with a strong emphasis on version control using Git. This ensured an organized coding environment, facilitating effective teamwork and streamlined project management. The experiential nature of this program aimed at providing a comprehensive understanding of Data Science, emphasizing the significance of staying abreast of emerging tools and methodologies in an ever-evolving data-driven landscape. The internship also introduced data science methodologies, instilling adaptability—an indispensable skill in dynamic real-world scenarios where continuous learning and innovation are crucial.

1. INTRODUCTION

Data Science encapsulates a dynamic and multifaceted discipline that revolves around extracting actionable insights from large and complex datasets. It entails the fusion of statistical analysis, machine learning, and domain expertise to decipher patterns and trends. Programming languages such as Python and R serve as the backbone, complemented by libraries like NumPy, Pandas, and Scikit-learn for data manipulation and analysis. Jupyter Notebooks provide an interactive platform for collaborative data exploration. Machine Learning, a core element, involves deploying frameworks like TensorFlow and PyTorch to build models for predictive analytics and decision-making. Visualization tools such as Matplotlib and Tableau facilitate the creation of meaningful visual representations. Effective database management, encompassing SQL, NoSQL, and technologies like MongoDB, ensures efficient data handling. Data Scientists also navigate ethical considerations, prioritizing responsible data use and privacy compliance. In this rapidly evolving field, staying abreast of emerging technologies is imperative, highlighting the need for continuous learning and adaptability to navigate the ever-changing landscape of data science tools and methodologies.

2. BACKGROUND

Data Science is an interdisciplinary field that combines statistical analysis, machine learning, and domain expertise to extract valuable insights from large datasets. A Data Scientist serves as a proficient navigator in this realm, possessing a diverse skill set that spans data acquisition, cleaning, exploration, modeling, and interpretation.

Data Scientist: A Data Scientist is well-versed in the entire data lifecycle, from data collection and preprocessing to model development and result interpretation.

Data Acquisition and Cleaning: Data Scientists are adept at collecting and cleaning diverse datasets, ensuring data quality and consistency. They utilize tools like Python and R for efficient data manipulation and cleansing.

Exploratory Data Analysis: This phase involves a comprehensive exploration of the dataset, including the identification of patterns, anomalies, and key trends. Visualization tools such as Matplotlib and Seaborn play a vital role in this process.

Machine Learning: Data Scientists employ various machine learning algorithms build predictive models. Frameworks like TensorFlow and PyTorch are utilized for tasks such as classification, regression, and clustering.

Data Visualization: Effective communication of findings is facilitated through data visualization. Data Scientists utilize tools like Tableau and Power BI to create insightful and impactful visualizations.

Database Management: Understanding databases is fundamental to Data Science. Data Scientists work with SQL and NoSQL databases, using technologies like MongoDB and MySQL for efficient data storage and retrieval.

Ethical Considerations: Data Scientists operate with a strong sense of ethics, ensuring responsible and secure use of data. They navigate privacy compliance and contribute to the ethical development and deployment of machine learning models.

Continuous Learning: In the rapidly evolving landscape of Data Science, staying updated on the latest algorithms, frameworks, and industry trends is crucial. Data Scientists exhibit a commitment to continuous learning and adaptability to tackle emerging challenges in the field.

3. OBJECTIVES

Comprehensive Data Exploration with Python and R: The primary objective is to delve into extensive data exploration using Python and R, conducting descriptive and inferential statistical analysis. This includes gaining proficiency in data manipulation libraries like Pandas and NumPy for efficient data handling.

Predictive Modeling using Machine Learning Frameworks: The goal is to develop expertise in constructing predictive models through machine learning frameworks such as TensorFlow and PyTorch. This involves implementing various algorithms for tasks like classification, regression, and clustering.

Effective Data Visualization with Matplotlib and Seaborn: The objective is to master the art of data communication through impactful visualizations using tools like Matplotlib and Seaborn. This includes creating plots, charts, and graphs to present insights in a clear and compelling manner.

Database Integration and SQL Mastery: The aim is to integrate databases seamlessly into the data science workflow, utilizing SQL for efficient data retrieval and manipulation. Data Scientists are expected to design well-structured database schemas, write optimized queries, and ensure data integrity.

Ethical Considerations and Responsible Data Handling: The objective involves instilling a strong sense of ethics in data science practices. Data Scientists are expected to navigate ethical considerations, ensuring privacy compliance and responsible use of data throughout the entire data science lifecycle.

Continuous Learning and Adaptability: In the dynamic landscape of data science, the goal is to foster a culture of continuous learning and adaptability. Data Scientists are encouraged to stay updated on emerging algorithms, frameworks, and industry trends, demonstrating a commitment to ongoing professional development.

Collaboration across Disciplines: The aim is to facilitate interdisciplinary collaboration, ensuring effective communication between data scientists, domainexperts, and other stakeholders. This involves integrating data science solutions seamlessly into the broader organizational context for impactful decision-making.

4. PURPOSE OF THE WORK

Technology Stack Exploration:

Aimed for a comprehensive learning experience by focusing on a technology stack including Python and R, along with key libraries like NumPy, Pandas, Matplotlib, and Seaborn in Data Science.

Python and R for Data Exploration:

Emphasized extensive data exploration using Python and R, enhancing skills in data manipulation, statistical analysis, and visualization to build a strong foundation for handling diverse datasets.

Machine Learning with TensorFlow and PyTorch:

Shifted focus to machine learning with TensorFlow and PyTorch, aiming to develop proficiency in constructing and deploying predictive models, covering tasks such as classification, regression, and clustering.

Database Integration and SQL for Data Retrieval:

Dedicated segment focused on seamless database integration, utilizing SQL for efficient data retrieval and manipulation, ensuring well-structured schemas and optimized queries.

Data Visualization Mastery:

Deepened understanding of data visualization principles, using tools like Matplotlib and Seaborn to create impactful visual representations for effective communication of findings.

Ethical Data Science Practices:

Emphasized ethical considerations, navigating privacy compliance, and ensuring responsible data handling throughout the internship, adhering to ethical standards in machine learning model development.

Continuous Learning and Collaboration:

Instilled a culture of continuous learning, encouraging Data Scientists to stay updated on emerging tools, while emphasizing collaborative practices for effective communication and integration of data science solutions within the organizational context.

5. CONCEPTS EXPLORED

The internship program was centered around overcoming key challenges in Data Science, with a technology stack that included EDA (Exploratory Data Analysis) processing, Power BI, and proficiency in data visualization techniques. The program focused on honing skills in various areas, including mastering EDA processing to uncover patterns and insights in raw data. Data Visualization and Interpretation became a key learning point, as I delved into Power BI for creating visually compelling and interactive dashboards to effectively communicate complex findings.

6. DETAILED DESCRIPTION OF LEARNING

WEEK 1: EXPLORATORY DATA ANALYSIS (EDA) BASICS

During the first week of the internship, the focus was on laying the foundation for data exploration and analysis through Exploratory Data Analysis (EDA). Key learning objectives and activities included:

Introduction to EDA:

Understanding the significance of EDA in the data science workflow.

Exploring the role of EDA in uncovering patterns, trends, and anomalies in datasets.

EDA Techniques:

Learning and applying basic statistical measures for data summarization.

Exploring graphical techniques for data visualization, such as histograms, box plots, and scatter plots.

Hands-On Practice:

Working with a sample dataset to apply EDA techniques.

Cleaning and preprocessing data to prepare it for analysis.

Data Interpretation:

Developing skills in interpreting EDA results.

Identifying potential insights and areas for further analysis.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Formatted Date	Summary	Precip Type	Temperature (C)	Apparent Temper	Humidity	Wind Speed (km/	Wind Bearing (	Visibility (km)	Loud Cover	Pressure (millibars)	Daily Summary
2	2006-04-01 00:00:00.000 +0200	Partly Cloudy	rain	9.472222222	7.388888889	0.89	14.1197	251	15.8263	0	1015.13	Partly cloudy throughout the day.
3	2006-04-01 01:00:00.000 +0200	Partly Cloudy	rain	9.355555556	7.227777778	0.86	14.2646	259	15.8263	0	1015.63	Partly cloudy throughout the day.
4	2006-04-01 02:00:00.000 +0200	Mostly Cloudy	rain	9.377777778	9.377777778	0.89	3.9284	204	14.9569	0	1015.94	Partly cloudy throughout the day.
5	2006-04-01 03:00:00.000 +0200	Partly Cloudy	rain	8.288888889	5.944444444	0.83	14.1036	269	15.8263	0	1016.41	Partly cloudy throughout the day.
6	2006-04-01 04:00:00.000 +0200	Mostly Cloudy	rain	8.755555556	6.977777778	0.83	11.0446	259	15.8263	0	1016.51	Partly cloudy throughout the day.
7	2006-04-01 05:00:00.000 +0200	Partly Cloudy	rain	9.222222222	7.111111111	0.85	13.9587	258	14.9569	0	1016.66	Partly cloudy throughout the day.
8	2006-04-01 06:00:00.000 +0200	Partly Cloudy	rain	7.733333333	5.522222222	0.95	12.3648	259	9.982	0	1016.72	Partly cloudy throughout the day.
9	2006-04-01 07:00:00.000 +0200	Partly Cloudy	rain	8.772222222	6.527777778	0.89	14.1519	260	9.982	0	1016.84	Partly cloudy throughout the day.
10	2006-04-01 08:00:00.000 +0200	Partly Cloudy	rain	10.82222222	10.82222222	0.82	11.3183	259	9.982	0	1017.37	Partly cloudy throughout the day.
11	2006-04-01 09:00:00.000 +0200	Partly Cloudy	rain	13.77222222	13.77222222	0.72	12.5258	279	9.982	0	1017.22	Partly cloudy throughout the day.
12	2006-04-01 10:00:00.000 +0200	Partly Cloudy	rain	16.01666667	16.01666667	0.67	17.5651	290	11.2056	0	1017.42	Partly cloudy throughout the day.
13	2006-04-01 11:00:00.000 +0200	Partly Cloudy	rain	17.14444444	17.14444444	0.54	19.7869	316	11.4471	0	1017.74	Partly cloudy throughout the day.
14	2006-04-01 12:00:00.000 +0200	Partly Cloudy	rain	17.8	17.8	0.55	21.9443	281	11.27	0	1017.59	Partly cloudy throughout the day.
15	2006-04-01 13:00:00.000 +0200	Partly Cloudy	rain	17.33333333	17.33333333	0.51	20.6885	289	11.27	0	1017.48	Partly cloudy throughout the day.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import matplotlib.pyplot as plt # Plotting library for graph
import seaborn as sns#Loading seaborn library
import string

# read the csv file to load the dataset
data = pd.read_csv('/content/drive/MyDrive/test.csv')
print(data)
data.head()
```

```
       Temperature (C)  Apparent Temperature (C)  Humidity  Wind Speed (km/h)  \
0             9.472222                  7.388889      0.89            14.1197
1             9.355556                  7.227778      0.86            14.2646
2             9.377778                  9.377778      0.89             3.9284
3             8.288889                  5.944444      0.83            14.1036
4             8.755556                  6.977778      0.83            11.0446
...                ...                       ...       ...                ...
96448        26.016667                 26.016667      0.43            10.9963
96449        24.583333                 24.583333      0.48            10.0947
96450        22.038889                 22.038889      0.56             8.9838
96451        21.522222                 21.522222      0.60            10.5294
96452        20.438889                 20.438889      0.61             5.8765
```

[96453 rows x 12 columns]

	Formatted Date	Summary	Precip Type	Temperature (C)	Apparent Temperature (C)	Humidity	Wind Speed (km/h)	Wind Bearing (degrees)	Visibility (km)	Loud Cover	Pressure (millibars)	Daily Summary
0	2006-04-01 00:00:00.000 +0200	Partly Cloudy	rain	9.472222	7.388889	0.89	14.1197	251.0	15.8263	0.0	1015.13	Partly cloudy throughout the day.
1	2006-04-01 01:00:00.000 +0200	Partly Cloudy	rain	9.355556	7.227778	0.86	14.2646	259.0	15.8263	0.0	1015.63	Partly cloudy throughout the day.
2	2006-04-01 02:00:00.000 +0200	Mostly Cloudy	rain	9.377778	9.377778	0.89	3.9284	204.0	14.9569	0.0	1015.94	Partly cloudy throughout the day.
3	2006-04-01 03:00:00.000 +0200	Partly Cloudy	rain	8.288889	5.944444	0.83	14.1036	269.0	15.8263	0.0	1016.41	Partly cloudy throughout the day.
4	2006-04-01 04:00:00.000 +0200	Mostly Cloudy	rain	8.755556	6.977778	0.83	11.0446	259.0	15.8263	0.0	1016.51	Partly cloudy throughout the day.

```
#Data manipulation libraries :
import numpy as np  #numpy
import pandas as pd  #pandas
import tensorflow as tf
#System libraries
import glob #The glob module finds all the pathnames matching a specified pattern according to the rules used by the Unix shell

#Map plotting
import folium #Interactive Maps viz
```

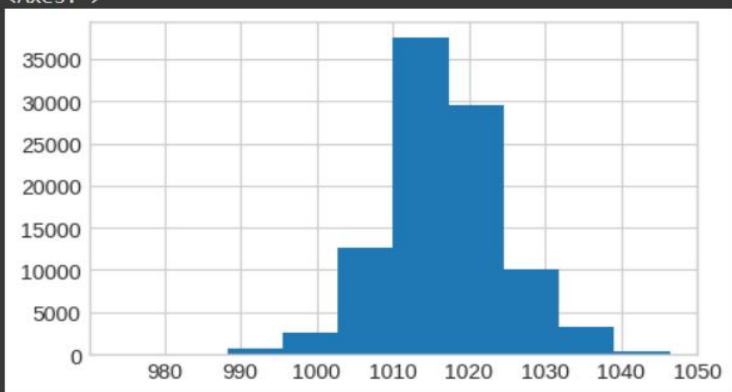
```
[ ]  data.dtypes

     Formatted Date              object
     Summary                     object
     Precip Type                 object
     Temperature (C)            float64
     Apparent Temperature (C)   float64
     Humidity                   float64
     Wind Speed (km/h)          float64
     Wind Bearing (degrees)     float64
     Visibility (km)            float64
     Loud Cover                 float64
     Pressure (millibars)       float64
     Daily Summary               object
     dtype: object
```

```
#Categorical variables:
categorical = data.select_dtypes(include = ["object"]).keys()
print(categorical)

Index(['Formatted Date', 'Summary', 'Precip Type', 'Daily Summary'], dtype='object')
```

```
[ ]  #Quantitative variables:
quantitative = data.select_dtypes(include = ["int64","float64"]).keys()
print(quantitative)

Index(['Temperature (C)', 'Apparent Temperature (C)', 'Humidity',
       'Wind Speed (km/h)', 'Wind Bearing (degrees)', 'Visibility (km)',
       'Loud Cover', 'Pressure (millibars)'],
      dtype='object')
```

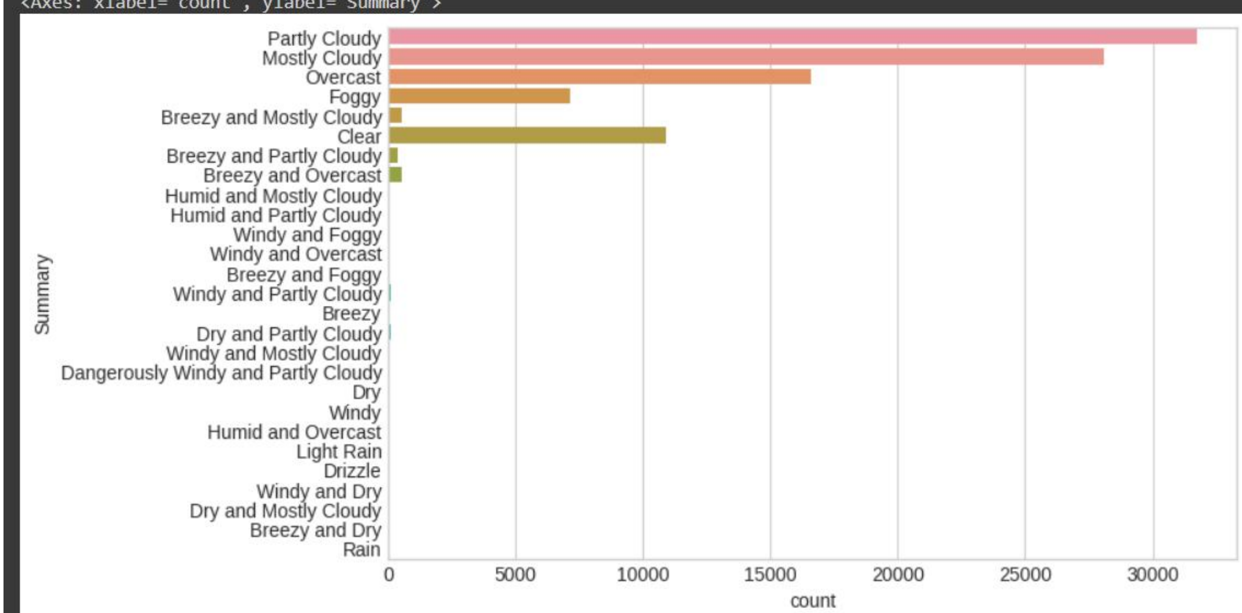
```python
pressure_median = data['Pressure (millibars)'].median()

def pressure(x):
    if x==0:
        return x + pressure_median
    else:
        return x

data["Pressure (millibars)"] = data.apply(lambda row:pressure(row["Pressure (millibars)"]) , axis = 1)

rcParams['figure.figsize'] = 5, 3
data['Pressure (millibars)'].hist()
```

```
<Axes: >
```




```python
rcParams['figure.figsize'] = 8, 5
sns.countplot(y=data['Summary'])
```

```
<Axes: xlabel='count', ylabel='Summary'>
```



WEEK 2: POWER BI LEARNING AND DASHBOARD CREATION

In the second week, the focus shifted to learning and utilizing Power BI for data visualization and dashboard creation. Key learning objectives and activities included:

Introduction to Power BI:

Understanding the role of Power BI in data visualization and reporting.

Exploring the Power BI interface and its key functionalities

Data Import and Transformation:

Learning to import data into Power BI from various sources.

Exploring data transformation and cleaning capabilities within Power BI.

Dashboard Design Principles:

Understanding best practices for designing effective dashboards.

Exploring the use of visual elements to enhance data storytelling.

Hands-On Dashboard Creation:

Applying Power BI skills to create a comprehensive dashboard.

Integrating insights gained from EDA into the dashboard.

Data Interaction and Sharing:

Exploring interactive features in Power BI for enhanced user engagement.

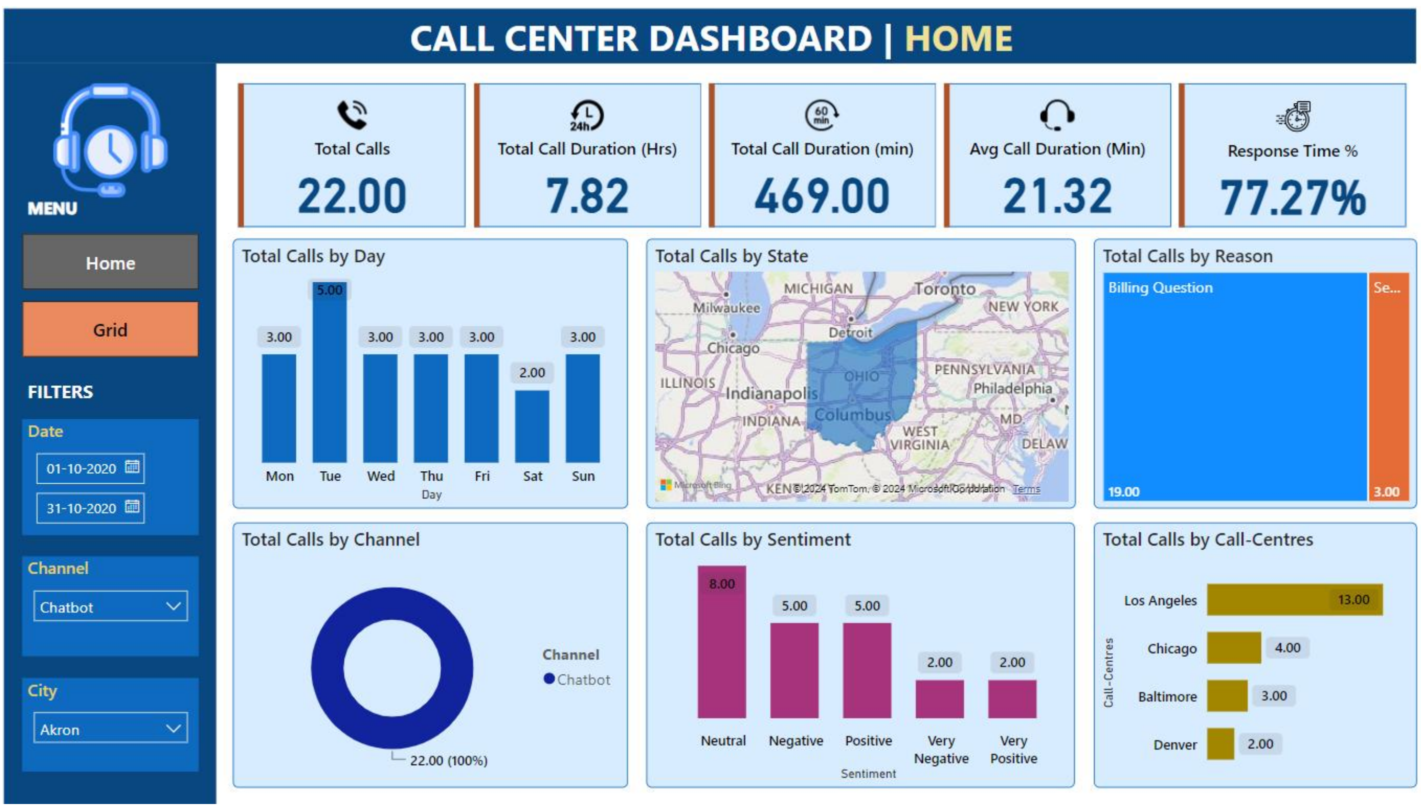
Learning to share and collaborate on Power BI dashboards.

These two weeks provided a solid foundation in both exploratory data analysis and data visualization using Power BI, setting the stage for continued learning and practical application in subsequent weeks of the internship.

Id	Call Timestam	Call-Centres City	Channel	City	Customer Name	Reason	Response Time	Sentiment	State	Call Duration In Minutes	Csat Score
DKK-57076809-w-055481-fU	29-10-2020	Los Angeles	Call-Center	Detroit	Analise Gairdner	Billing Question	Within SLA	Neutral	Michigan	17	7
QGK-72219678-w-102139-KY	05-10-2020	Baltimore	Chatbot	Spartanburg	Crichton Kidsley	Service Outage	Within SLA	Very Positive	South Carolina	23	
GYJ-30025932-A-023015-LD	04-10-2020	Los Angeles	Call-Center	Gainesville	Averill Brundrett	Billing Question	Above SLA	Negative	Florida	45	
ZJI-96807559-i-620008-m7	17-10-2020	Los Angeles	Chatbot	Portland	Noreen Lafflina	Billing Question	Within SLA	Very Negative	Oregon	12	1
DDU-69451719-O-176482-Fm	17-10-2020	Los Angeles	Call-Center	Fort Wayne	Toma Van der Beken	Payments	Within SLA	Very Positive	Indiana	23	
JVI-79728660-U-224285-4a	28-10-2020	Baltimore	Call-Center	Salt Lake City	Kaylyn Emlen	Billing Question	Within SLA	Neutral	Utah	25	5
AZI-95054097-e-185542-PT	16-10-2020	Baltimore	Chatbot	Tyler	Phillipe Bowring	Billing Question	Within SLA	Neutral	Texas	31	8
TWX-27007918-I-608789-Xw	21-10-2020	Los Angeles	Chatbot	New York City	Krysta de Tocqueville	Billing Question	Below SLA	Positive	New York	37	
XNG-44599118-P-344473-ZU	03-10-2020	Baltimore	Email	Dallas	Oran Lifsey	Billing Question	Below SLA	Very Negative	Texas	37	
RLC-64108207-Z-285141-VS	07-10-2020	Baltimore	Chatbot	Cincinnati	Port Inggall	Billing Question	Within SLA	Neutral	Ohio	12	
RJF-00263922-O-647027-TB	09-10-2020	Los Angeles	Chatbot	Everett	Ella Cristoforo	Billing Question	Within SLA	Negative	Washington	35	
ZQN-32874873-e-786499-kJ	11-10-2020	Los Angeles	Web	Huntington	Aubrey Surcombe	Billing Question	Within SLA	Negative	West Virginia	18	
JDP-35147568-w-630120-3l	02-10-2020	Baltimore	Call-Center	Portland	Nicolle Fareweather	Billing Question	Within SLA	Very Positive	Oregon	30	
DPT-56483482-P-371409-CQ	10-10-2020	Denver	Chatbot	Springfield	Melesa Ricardot	Billing Question	Within SLA	Positive	Massachusetts	20	7

DASHBOARD:

HOME VIEW:



GRID VIEW:

7. RESULTS AND DISCUSSION:

The internship yielded significant achievements in the realm of data science, particularly in the areas of EDA processing, Power BI utilization, dashboard creation, and data visualization. The proficiency in EDA processing was demonstrated through effective data cleaning and processing, revealing insights and patterns in diverse datasets. Mastery in Power BI allowed for the creation of insightful visualizations and interactive dashboards, enhancing the ability to present complex data in an understandable and visually appealing manner.

In the context of data science technologies, the internship emphasized the development of robust skills in EDA processing, showcasing the ability to prepare and clean data for analysis. Dedicated efforts were directed towards Power BI, resulting in the creation of meaningful and visually engaging dashboards. Additionally, the internship provided practical insights into collaborative development practices, fostering effective communication and integration within data science and analytics teams.

The dual exposure to both EDA processing and Power BI utilization proved integralto a comprehensive understanding of data science. Emphasis on version control, utilizing tools like Git, maintained an organized and collaborative coding environment, contributing to streamlined project management and clear documentation of the data analysis process. The internship effectively equipped me with the skills and knowledge required for successful contributions in the dynamic field of data science.

8. CONCLUSION AND FUTURE ENHANCEMENTS

The internship report encapsulates my journey in a data science internship, focusing on essential concepts such as EDA (Exploratory Data Analysis) processing, Power BI learning, dashboard creation, and data visualization. I immersed myself in the realm of data cleaning and processing, utilizing tools and techniques to unveil patterns and insights within raw datasets. Proficiency in Power BI allowed me to transform complex data into visually engaging dashboards, enhancing my ability to communicate meaningful findings effectively.

On the technical front, I delved into the intricacies of EDA processing, honing skills in preparing and cleaning data for analysis. The learning curve extended to Power BI, where I mastered the art of creating insightful visualizations and interactive dashboards. This experience solidified my foundation in data science, exposing me to industry-standard tools and collaborative practices prevalent in the field.

The hands-on application of skills in constructing meaningful visualizations and dashboards proved invaluable for my data science career. The internship effectively provided comprehensive training in key aspects of data science, fostering proficiency in EDA processing, Power BI, and data visualization techniques. This exposure equips me with the adaptability required for real-world data-driven scenarios, positioning me for continued growth and success in the dynamic field of data science.

9. REFERENCES

https://learn.microsoft.com/en-us/power-bi/fundamentals/service-get-started
https://www.tutorialspoint.com/power_bi/index.htm
https://www.analyticsvidhya.com/blog/2022/07/step-by-step-exploratory-data-analysis-eda-using-python/
https://www.geeksforgeeks.org/exploratory-data-analysis-in-python/